

End-to-End Automated Bill of Materials & Supplier Procurement Process

Document Type	Version	Date	Classification
			Confidential

Executive Summary

This document describes the full end-to-end BOM Procurement Workflow — a six-stage automated system that transforms client order and specification inputs into a production-ready material confirmation. The workflow integrates AI-driven document analysis, live inventory reconciliation, and automated supplier communication to eliminate manual bottlenecks and accelerate production release timelines.

Scope of this Document

This specification covers the complete operational workflow from MD document submission through to production queue release. It is intended for review by senior management and system stakeholders to align on process design, approval authorities, and automation boundaries.

Workflow at a Glance

Stage	Name	Owner	Automation Level	Key Output
1	MD/Direct Manager Submission	MD/Direct Manager	Manual Upload	Order + Spec Files
2	AI Analysis & BOM Gen.	AI Engine	Fully Automated	Draft BOM Table
3	MD Review & Approval	Managing Director	Human Decision	Approved BOM
4	Inventory Check	System	Fully Automated	Stock Status Report

5	Supplier Call & Procurement	System	Fully Automated	Confirmed Quantities
6	Completion & Prod. Release	System / MD	Automated Alert	Production Queue Entry

STAGE 1

MD/Direct Manager Submission

Secure document upload initiating the procurement cycle

The workflow is initiated when the MD/Direct Manager authenticates and logs into the application. Upon access, the MD/Direct Manager is presented with a dedicated upload screen designed to accept two mandatory documents simultaneously.

Key Inputs & Functions

Order Sheet	Contains item quantities, SKUs (Stock Keeping Unit), and delivery dates required for fulfillment.
Specification Sheet	Contains physical and technical parameters including dimensions, materials, and tolerances.

Process Steps

1	MD/Direct Manager logs in through the secure application portal.
2	The upload interface is presented with clearly labelled fields for each document type.
3	Both documents are submitted through the in-app upload mechanism.
4	Upon successful submission, both files are immediately passed to the AI engine for processing.
5	The system confirms receipt and initiates Stage 2 automatically.

Document Integrity

Both documents must be submitted together in a single transaction. The system will not proceed to AI analysis until both files are present and validated for format compliance. Accepted formats include PDF, XLSX, and CSV.

STAGE 2

AI Analysis and BOM Generation

Automated extraction and structured BOM compilation from submitted documents

Once both documents are received, the AI engine begins parallel analysis. It performs cross-document reconciliation — reading the Order Sheet for procurement quantities and the Specification Sheet for material requirements — then compiles these into a unified, structured Bill of Materials.

Key Inputs & Functions

Data Extraction	The AI extracts item codes, descriptions, quantities, material, color, sizes, thickness as well as comments from the Order Sheet.
Specification Mapping	Specification requirements are mapped to corresponding material and component needs from the Spec Sheet.

Process Steps

1	AI engine reads the Order Sheet and identifies all line items, SKUs (Stock Keeping Unit), quantities, and delivery dates.
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2	AI engine reads the Specification Sheet and extracts material grades, dimensions, and tolerances per item.
3	Cross-document mapping is performed to align each order line with its specification requirements.
4	A structured, editable BOM table is generated with all extracted fields populated and a single BOM for a single style. And there's only one BOM per style and one for an order as well.
5	The draft BOM is presented to the Managing Director in the review interface.

Output Format

The AI-generated BOM is presented as a structured, editable table. Fields include: Item Code, Description, Quantity Required, Unit of Measure, Material Specification, Tolerance Class and Price. All fields are pre-populated from the source documents and remain fully editable by the MD prior to approval.

Source Documents Used in AI Analysis

The AI engine processes two primary source documents at Stage 2: the **Order Sheet** and the **Specification Sheet** (issued by Beau Geste / the buyer). Both documents are submitted simultaneously at Stage 1 and are available to the AI engine as soon as Stage 2 begins. The actual documents used for this production run are reproduced below for reference.

Order Sheet — Beau Geste (No. 1579)

This is the buyer-issued production order from Beau Geste (株式会社ボージェスト) confirming a bulk production run for CRIMIE brand leather trousers. The AI engine extracts the following key fields: manufacturer (PAKKAR), article name (Leather Pants), material (Sheep skin / Sheep Glass), customer reference (CR1-02F5-PL02), internal reference (A32073-44), total quantity (60 pcs), size breakdown (S:4, M:17, L:23, XL:10, XXL:6), and the unit price (\$104). The handwritten note on the order also instructs special attention to the placement of goat suede panels, which the AI flags as a specification annotation.

発注書
Order Sheet

No. 1579

発注先 Manufacturer
PAKKAR

発注日 Date
1 / Jun / 2026

Beau Geste
株式会社ボージェスト

出荷期限 Shipping Limit	色 Color	合計数量 Amount	単価 Unit Price	合計単価 Total Price
/	Black	60 pcs	\$ 104	

24.5 of (Sheep Glass)
2.6 of (Goat Suede)

品名 Article Name	素材 Material
Leather Pants	Sheep skin

当社管理番号 Our Ref-No	製品番号 Customer No
A32073-44	CR1-02F5-PL02

Skin Order Sheet No.

サイズ Size					
S	4				
M	17				
L	23				
XL	10				
XXL	6				

Approval Sample

仕様
Specification

Bulk Production

• Article: Sheep Glass
• Thickness: 0.6~0.7 m/m

pls take care of where it is used goat suede.

備考
Note

Customer: CRIMIE

起票 S.Aiya
担当 S.Aiya
承認

ネーム 下り札 附属 バターン

BOM / Price Quotation Sheet — Pakkar Tanveer Exports

This is the factory-prepared Yardage and Price Quotation for style CRI 02F5 PL02 (CRIMIE brand), produced by Pakkar Tanveer Exports. The AI engine reads this document to populate the Bill of Materials with the following detail: main materials (Sheep Glass Col Black — 34.5 dm² at \$1.80/unit, and Goat Suede — 2.6 dm² at \$1.75/unit), sub-materials (Polyster Lining for pockets — 0.4 dm² at \$1.50/unit, Fusing Interlining — 1 unit at \$1.00, Thread — 1 unit at \$1.00), accessories (Buttons SGT-10 20mm — qty 4, Buttons SGT-10 15mm — qty 2, Support Buttons 10mm — qty 6), and cost components (Cutting and Stitching \$30.00, Packaging \$3.00, FOB Charge / Buyer Material Duty and Clearing Charges \$5.00). The total FOB garment price extracted is US\$107.25. The following are the columns of BOM sheet: No. or Name, Yardage, Unit, Price and Mass. The following are the rows of BOM sheet: Main material, Sub-Material(lining, etc), Threads, Accessories, Manufacturing cost, Packaging cost, FOB(Freight On Board) charge and also Total.

Yardage and Price Quotation					
Brand	CRIMIE				
Style No.	CRI 02F5 PL02				
Domestic Price					
Garment Price	US\$107.25				
Quotation: FOB					
Factory	PAKKAR TANVEER EXPORTS				



	No. or Name	Yardage	Unit	Sample	Mass
Main Material	SHEEP GLASS COL BLACK	34.5	1.80	62.1	
	GOAT SUEDE	2.6	1.75	4.55	
Sub-Material					
LINING	POLYSTER LINING - POCKETS	0.4	1.5	0.6	
INTERLINING	FUSING INTER LINING	1		1.00	
THREADS	THREAD	1		1.00	
ACCESSORIES	BUTTONS - SGT-10 20MM	4	0	0.00	
	BUTTONS - SGT-10 15MM	2	0	0.00	
	SUPPORT BUTTONS 10MM	6	0	0.00	
Manufacturing cost	CUTTING AND STITCHING	1	1	30.00	
Packaging cost	BOX, POLY COVER, TISSUE PAPER	1	1	3.00	
FOB CHARGE	BUYER MATERIAL DUTY + CLEARING CHARGES	1	1	5.00	
TOTAL				\$ 107.25	

AI Cross-Document Reconciliation Logic

Once both documents are ingested, the AI engine performs a structured reconciliation across four dimensions: (1) **Style / Reference Matching** — the customer reference CR1-02F5-PL02 on the Order Sheet is cross-matched against style CRI 02F5 PL02 on the BOM sheet to confirm both documents relate to the same garment; (2) **Quantity Propagation** — the total order quantity (60 pcs) is multiplied against each material yardage figure to calculate bulk material requirements (e.g. 34.5 dm²/garment x 60 = 2,070 dm² of Sheep Glass required); (3) **Specification Annotation** — freeform notes on the Order Sheet (such as the handwritten instruction regarding goat suede panel placement, and the material thickness specification of 0.6–0.7 m/m) are extracted and appended as annotations on the relevant BOM line items; (4) **Price Capture** — unit prices for all materials and cost components are captured from the BOM sheet and recorded against each line item in the draft BOM, providing cost visibility at the point of MD/DIRECT MANAGER review.

Category	Item Description	Qty/Garment	Bulk Qty (60 pcs)	Unit Price	Total Cost
Main Material	Sheep Glass Col Black	34.5 dm ²	2,070 dm ²	\$1.80/dm ²	\$3,726.00
Main Material	Goat Suede	2.6 dm ²	156 dm ²	\$1.75/dm ²	\$273.00

Category	Item Description	Qty/Garment	Bulk Qty (60 pcs)	Unit Price	Total Cost
Sub-Material (Lining)	Polyster Lining — Pockets	0.4 dm²	24 dm²	\$1.50/dm²	\$36.00
Interlining	Fusing Interlining	1 pc	60 pcs	\$1.00/pc	\$60.00
Threads	Thread	1 pc	60 pcs	\$1.00/pc	\$60.00
Accessories	Buttons SGT-10 20mm / 15mm, Support Buttons 10mm	12 pcs	720 pcs	TBC	TBC
Manufacturing	Cutting and Stitching	1	60	\$30.00	\$1,800.00
Packaging	Box, Poly Cover, Tissue Paper	1	60	\$3.00	\$180.00
FOB Charge	Buyer Material Duty + Clearing Charges	1	60	\$5.00	\$300.00
TOTAL FOB (60 pcs)					\$6,435.00

STAGE 3

MD Review and Approval

Human oversight gate — review, edit, and formal BOM approval

The Managing Director receives a notification and opens the AI-generated BOM in the review interface. This stage represents the primary human decision point in the workflow, where the MD exercises full editorial control over the BOM before committing it for procurement action.

Key Inputs & Functions

Edit Capability	The MD can adjust quantities, swap materials, or add and remove line items as required.
Approval Action	Once satisfied with the BOM content, the MD formally approves the document.

Process Steps

1	MD receives an in-app notification and automail within 5 hours of no response and an autocall within 5 hours of no response to the mail that the draft BOM is awaiting review.
2	MD opens the BOM in the review interface and inspects all line items.
3	Adjustments are made as required — modifying quantities, specifications, or line items.
4	MD approves the BOM via the formal approval action.
5	Approval locks the document, preventing further edits, and automatically triggers the inventory check.

Approval Lock

Upon MD approval, the BOM is cryptographically locked. No further modifications can be made to the approved version. A full revision history is maintained, including all edits made during the review period, the approving user identity, and the timestamp of approval.

STAGE 4

Inventory Check*Real-time stock reconciliation against approved BOM requirements*

With the BOM approved and locked, the system automatically initiates a live inventory check. Each line item in the approved BOM is cross-referenced against current warehouse stock levels. The system calculates the exact shortfall for any item that cannot be fulfilled from existing stock.

Key Inputs & Functions

Stock Comparison	Required quantity per BOM line is compared against quantity currently on hand in the warehouse system.
Status Assignment	Each line is assigned a status — SUFFICIENT, PARTIAL, or OUT OF STOCK — based on availability.

Process Steps

1	Approved BOM is passed to the inventory management module.
2	For every line item, the system queries live stock levels.
3	A status is assigned : SUFFICIENT (full quantity available), PARTIAL (some stock exists but shortfall present), or OUT OF STOCK (zero stock available).
4	For SUFFICIENT, PARTIAL, BALANCE & OUT OF STOCK items, the exact quantity is calculated.
5	All shortfall items are compiled into a procurement list and passed to Stage 5.

Status Definitions

SUFFICIENT — Full required quantity is available in stock; no procurement action needed. PARTIAL — Some stock exists but is insufficient; the shortfall quantity is forwarded for procurement. OUT OF STOCK — Zero units available; the full required quantity is forwarded for procurement.

STAGE 5

PO-Based Supplier Communication with Escalation*Email-driven supplier communication with fallback automated calling*

For every item identified as having a shortfall, the system generates a Purchase Order (PO) form and sends it to the designated supplier via email instead of initiating an immediate automated call. If the supplier does not respond within 5 hours of the email being sent, the system triggers an automated outbound call to follow up and obtain confirmation.

Key Inputs & Functions

Supplier Lookup	The system identifies the designated supplier for each shortfall item based on the article code.
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PO Generation & Email Dispatch	A PO form is automatically created and sent to the supplier's registered email address with material requirements.
Response Monitoring	The system tracks supplier response status within a defined 5-hour window.
Escalation via Call Execution	If no response is received within the timeframe, an automated call is placed to the supplier to communicate requirements and capture confirmation.

Process Steps

1	Identify shortfall item.
2	Map supplier using article code.
3	Generate PO form.
4	Send PO via email
5	Wait for supplier response (5 hours).
6	If no response → trigger automated call.
7	Record supplier confirmation.

Tools Used For Tracking :

Use tools like Mailtrack, HubSpot, Streak, or Yesware — these add a tiny invisible pixel to your email, and notify you when the recipient opens it.

Gmail users can install the Mailtrack Chrome extension for free read receipts.

KHALITH BAI									
PTE									
Po Form									
Po No	PO-DCA								
Total Order Qty	In Dcm	26710	In Sqft	2875	In Pcs		Date	16-05-2026	
SLNo	Article	Colour	Size	Substance	Selection	DCM	Price (DCM)	Priority	Delivery Date
1	SHEEP ANTIC	BLACK	4 TO 7	.07/08	1,2,3	26710	7/		23-05-2026
2									
3									
						26710			

NO VINE MARK AND BONE MARK
SKIN SIZE 4/7 (45 TO 50)

NOTE :

- Please notify us immediately if you are unable to delivery as specified.
- All test given copy should pass if fails in any lot you have to pay the compensation of garment.
- Payment terms : 90 days from the date as of whole lot of PO arrived.

Prepared by (Sender)

Authorised by

STAGE 6

Completion and Production Release
Final material-readiness confirmation and queue handoff

Once every line item in the approved BOM has been either fulfilled from existing warehouse stock or confirmed by a supplier through the automated response, the system evaluates overall material readiness and executes the final stage actions to release the job to production.

Key Inputs & Functions

Readiness Assessment	The system verifies that 100% of BOM line items are covered — either by stock on hand or confirmed supplier delivery.
Production Release	The job is marked as material-ready, an MD alert is triggered, and the job enters the production scheduling queue.

Process Steps

1	System verifies all BOM line items have either SUFFICIENT stock or confirmed supplier quantities.
2	Overall material readiness status is set to READY.
3	The MD receives an automated alert confirming material readiness.
4	The job is formally released to the production scheduling queue.
5	All procurement records, supplier confirmations, and inventory data are archived against the job record.

Production Queue Entry

The production scheduling queue entry includes: the approved BOM reference, confirmed material availability dates, supplier delivery schedules, and the original client delivery dates from the Order Sheet. This allows production planning to schedule manufacturing operations with full material visibility.

Roles and Responsibilities

Role	Stage Involvement	Responsibilities
MD/Direct Manager	Stage 1	Upload Order Sheet and Specification Sheet accurately and in full.
AI Engine	Stage 2	Parse submitted documents, extract BOM data, map specifications, generate structured draft BOM.
Managing Director	Stages 2, 3, 6	Review AI-generated BOM, make amendments, formally approve, receive material-ready alerts.
Inventory System	Stage 4	Perform live stock comparison, assign statuses, calculate shortfalls.
Procurement System	Stage 5	Look up suppliers, execute automated calls, record confirmations.
Production System	Stage 6	Receive released job, schedule manufacturing based on material availability.